

30P30N static overset verification

Problem: three distinct solid profiles (slat, main, flap), normalized chord.
One Gmsh fitted component contains all three physical walls and fluid slots;
a Cartesian background supplies the far field. This is a two-grid case,
not three independently moving fitted grids.

Equation: $-\text{Laplacian}(u) = -4$.

Exact field: $u = 1 + x^2 + y^2 + 0.2xy + 0.1x + 0.15y$.

Exact Dirichlet data apply only at physical walls and far-field boundaries.

Fringe values are unknowns constrained by ACTIVE donor interpolation.

Nonorthogonal FV fluxes and least-squares gradients are solved together.

Acceptance: both component volume-weighted L2 errors < 3%;
unique physical-domain integrated flux error < 3%; algebraic residual
and actual donor constraints < $1e-10$. Independent NumPy replay checks
raw polygons, source profiles, gradients, fluxes and donor constraints.

Scope: scalar equation verification. No overset pressure, velocity, C_p ,
lift/drag or turbulence validation is claimed. Point interpolation does
not provide strict local conservative interface flux coupling.

Geometry source: [linuxguy123/30P-30N-Validation-Case](#) (GPLv3).

Packaged profile files and exact numerical producer hashes are archived.

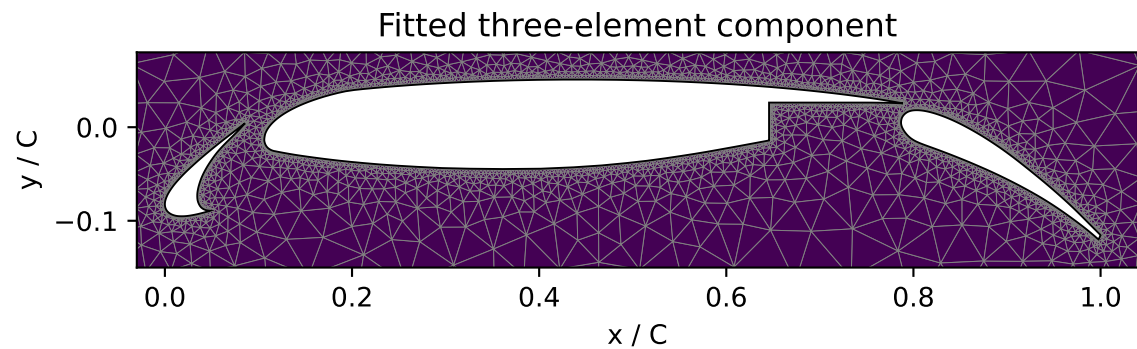
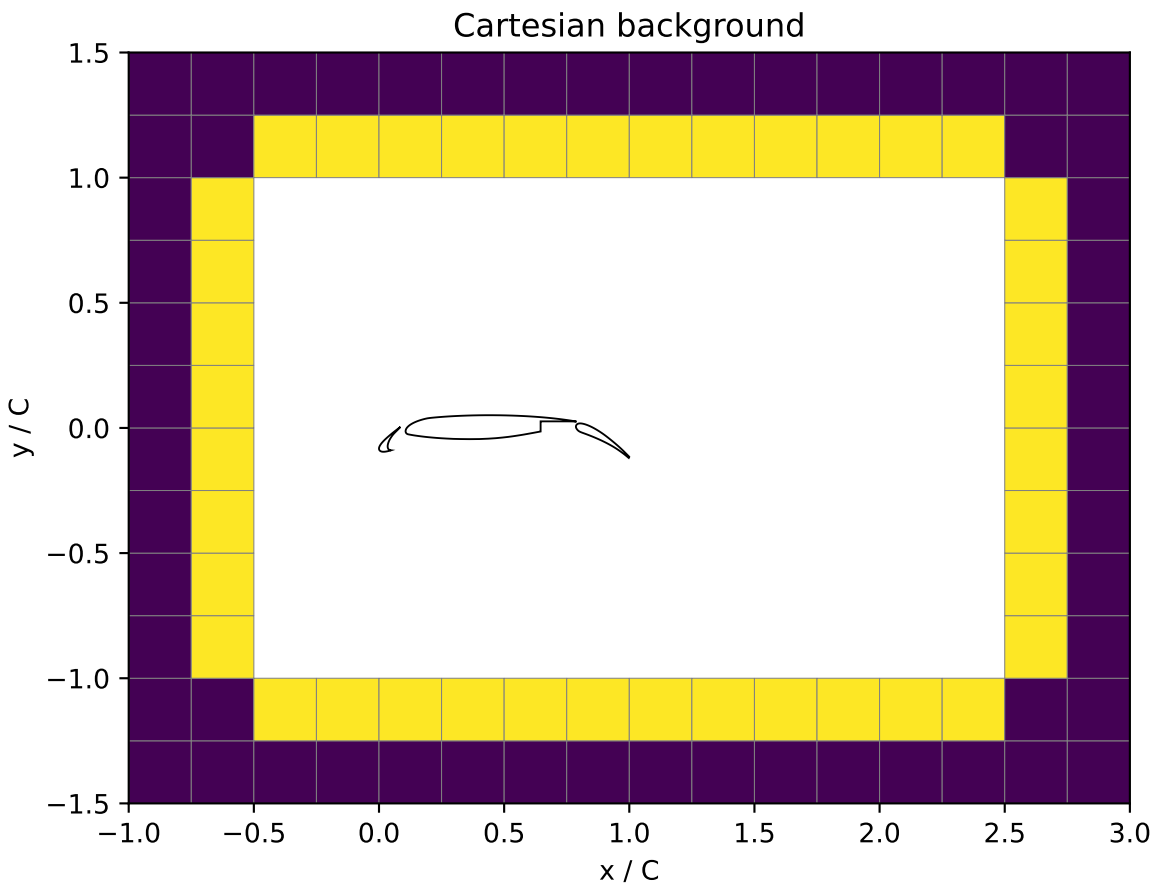
Reproduce from the repository:

```
PYTHONPATH=src python scripts/run_three_element_overset.py --output NEW_DIR
```

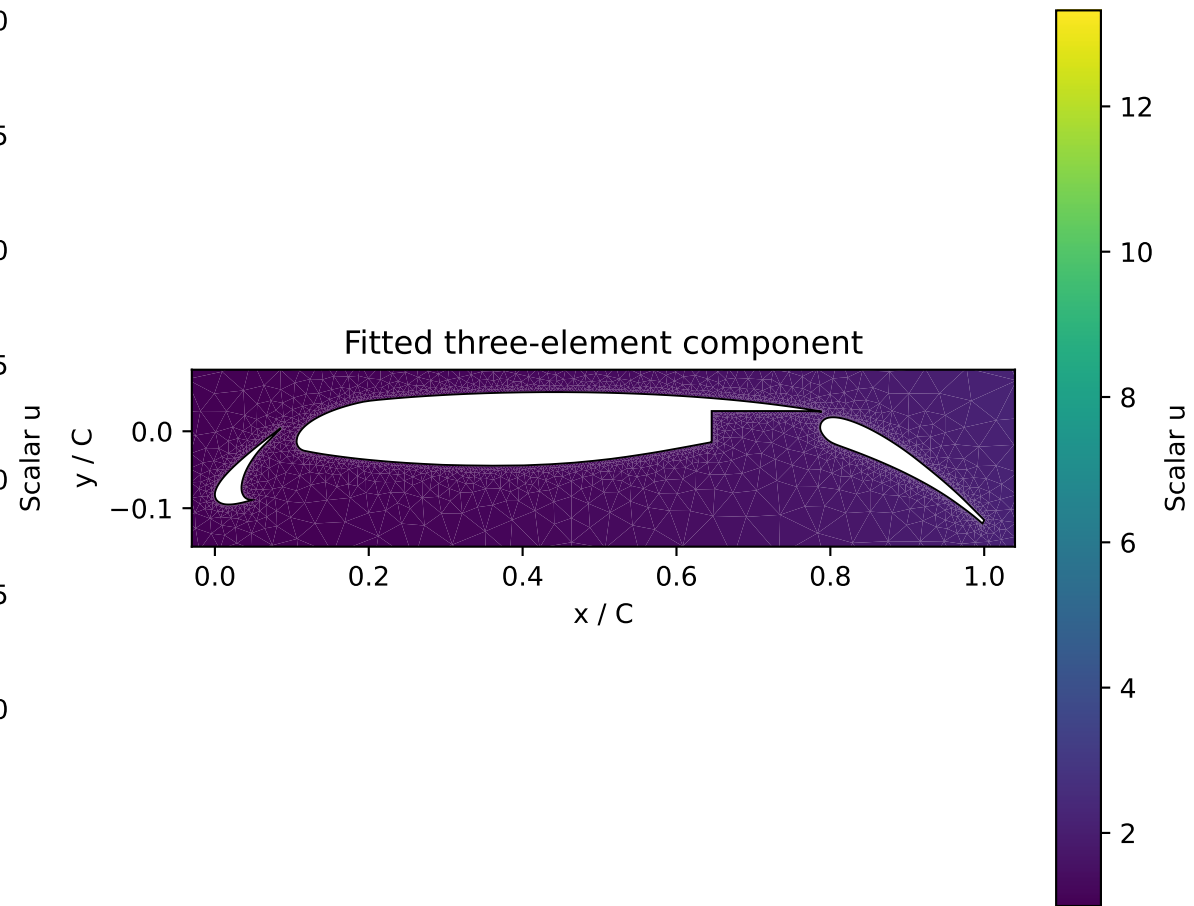
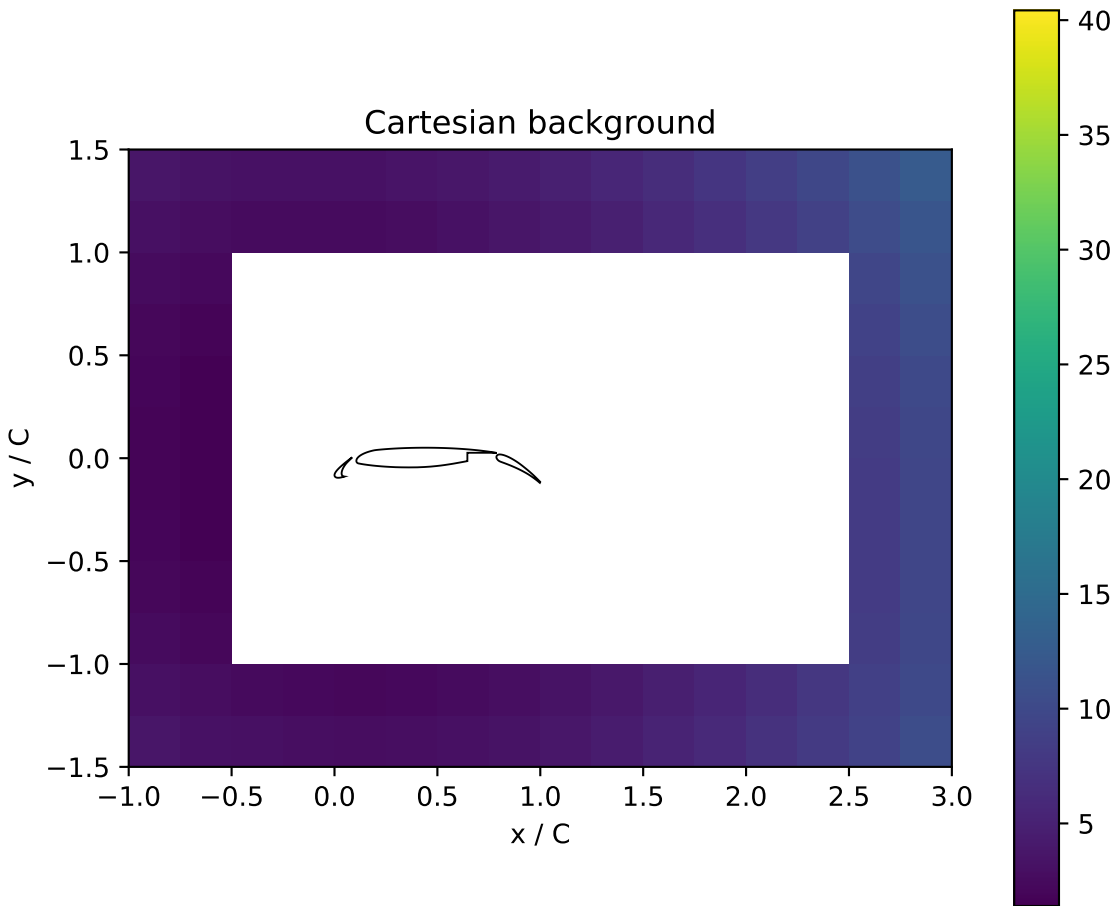
```
PYTHONPATH=src python scripts/audit_three_element_overset.py NEW_DIR
```

```
PYTHONPATH=src python scripts/report_three_element_overset.py NEW_DIR
```

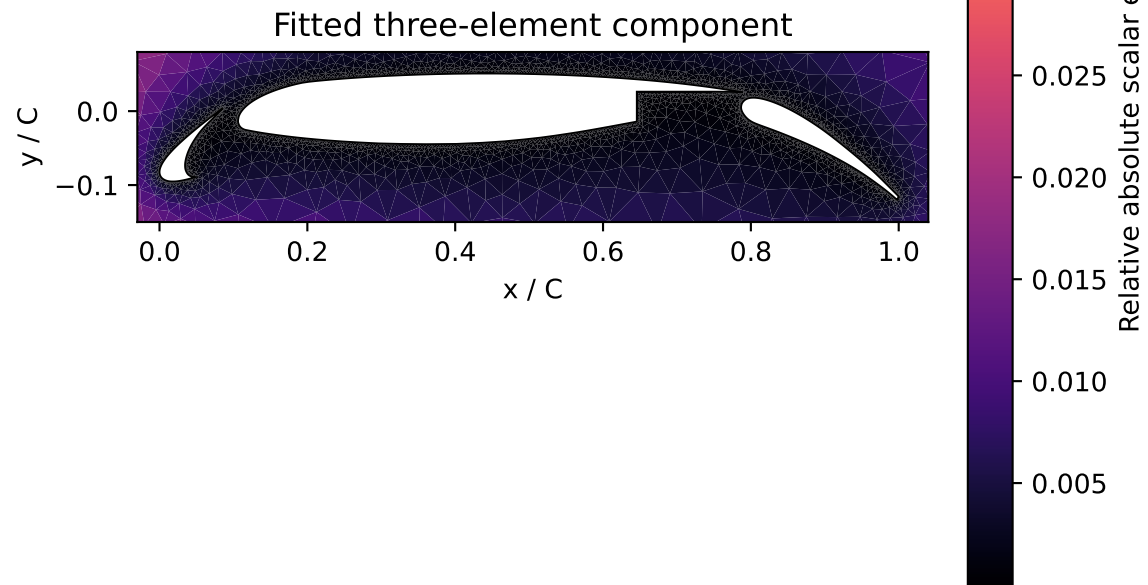
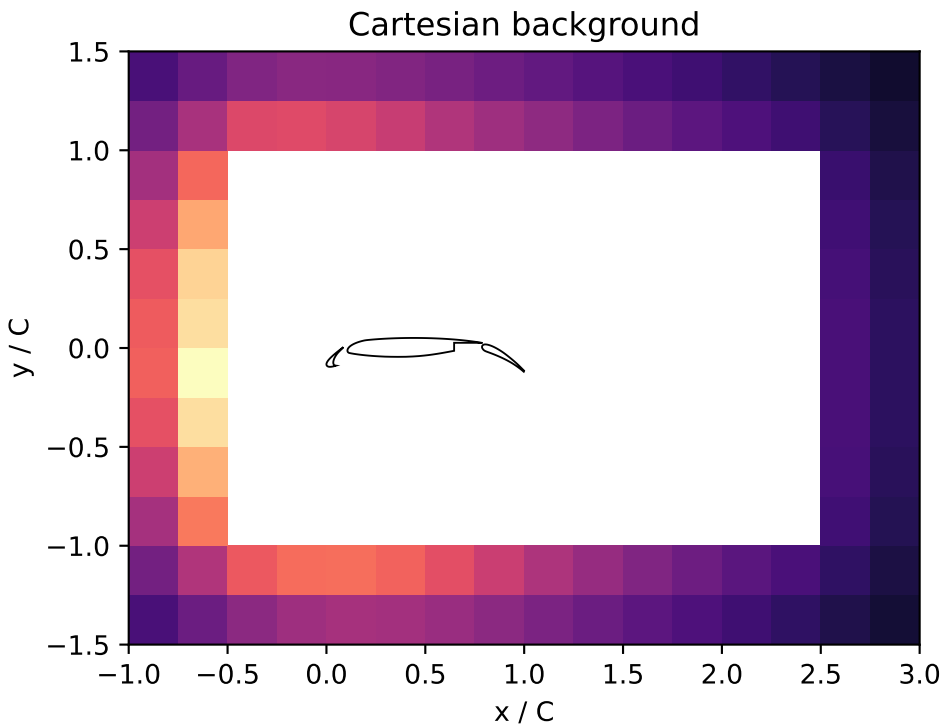
Background 32: mesh (physical-slot close-up)

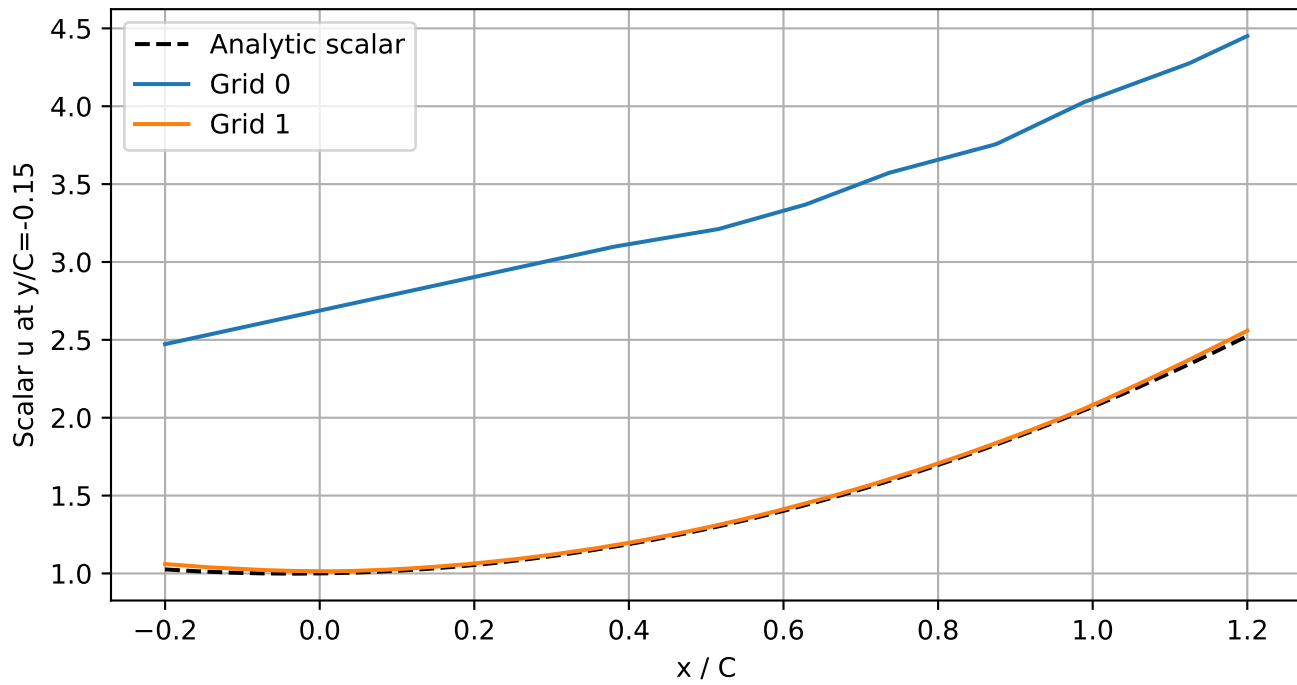


Background 32: solution (physical-slot close-up)



Background 32: error (physical-slot close-up)

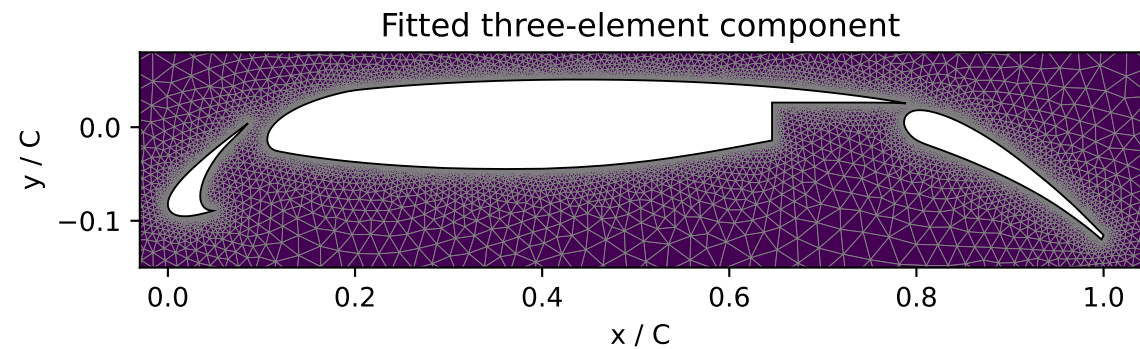
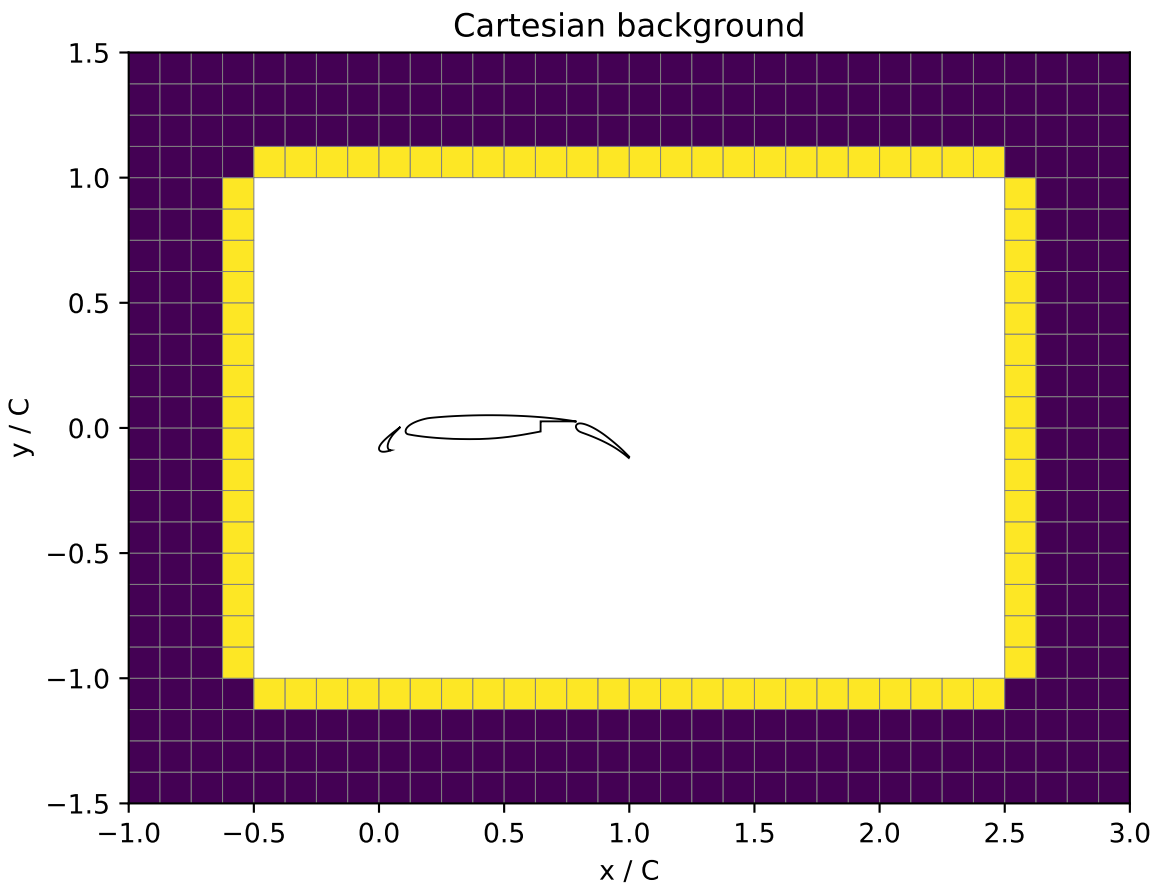




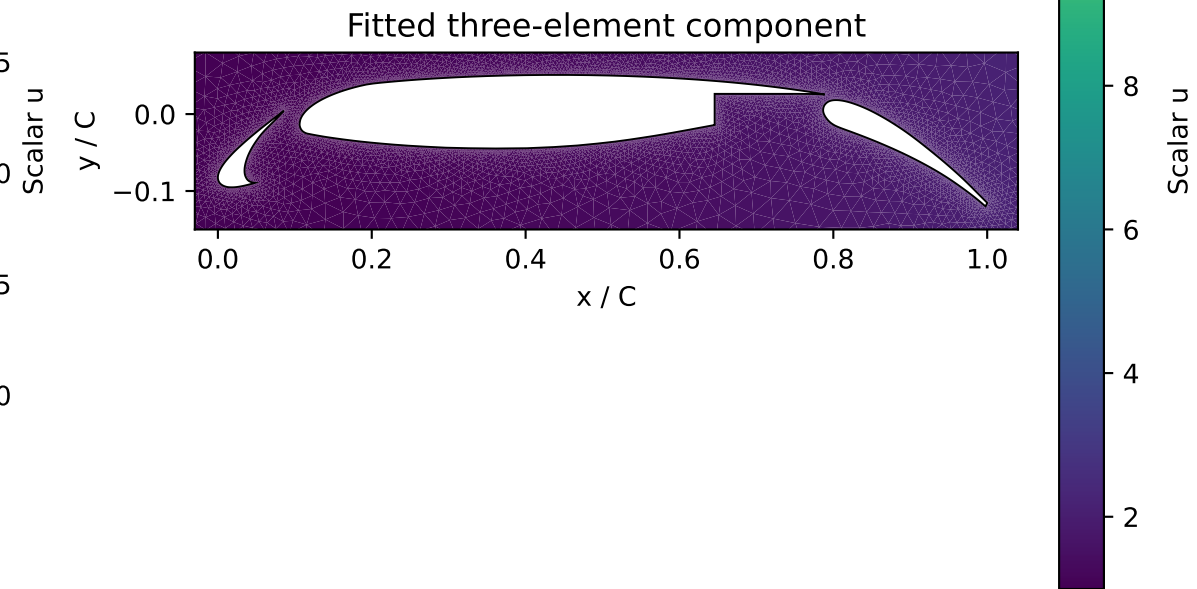
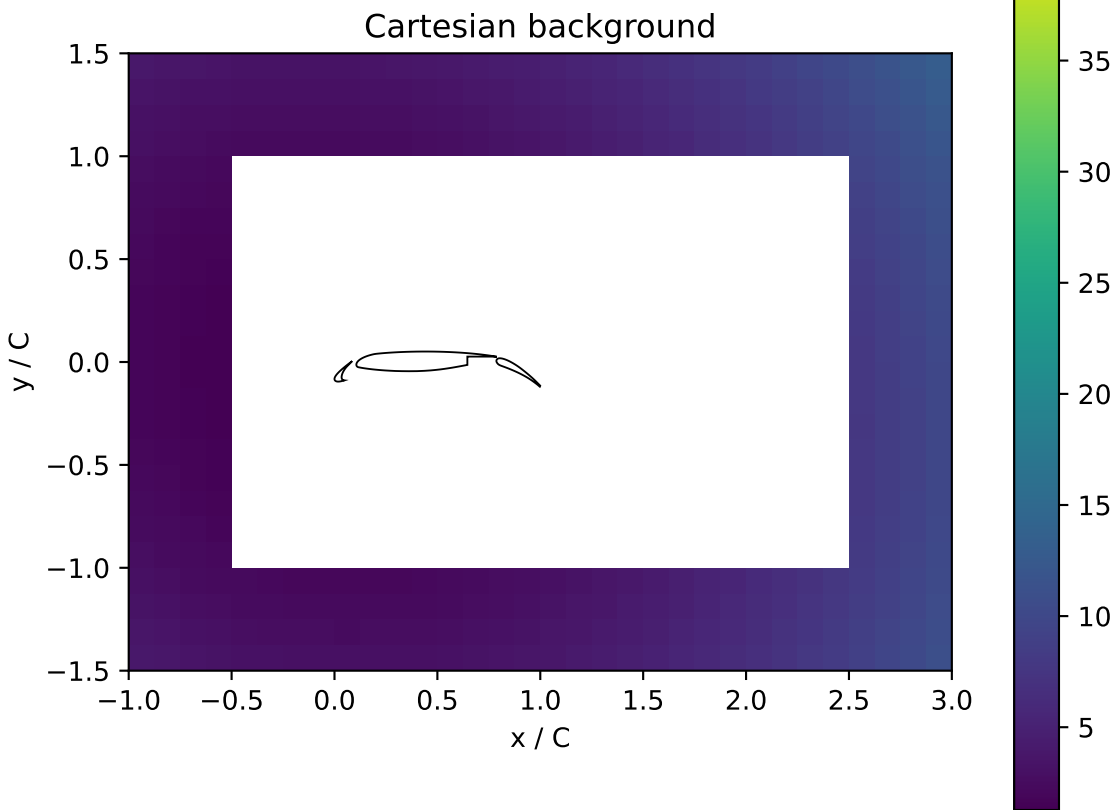
Grid 32: measured results

```
{
  "cells": [
    {
      "name": "background",
      "cells": 896,
      "active": 760,
      "hole": 96,
      "fringe": 40,
      "volume": 56.0
    },
    {
      "name": "30P30N fitted component",
      "cells": 12492,
      "active": 12374,
      "hole": 0,
      "fringe": 118,
      "volume": 11.948283660293896
    }
  ],
  "metrics": {
    "relative_algebraic_residual": 5.473854452463646e-16,
    "strict_local_conservation": false,
    "navier_stokes_verified": false,
    "grid_0_relative_l2_error": 0.001757311806124487,
    "grid_1_relative_l2_error": 0.013373023323663687,
    "maximum_donor_constraint_residual": 7.105427357601002e-15,
    "physical_global_conservation_defect_relative": 0.003980598476246738,
    "physical_boundary_diffusive_flux": -222.90230403042844,
    "physical_domain_integrated_source": -223.79313464117558,
    "active_cells": 13134,
    "fringe_cells": 158
  },
  "audit": {
    "resolution": 32,
    "passed": true,
    "independent_equation_max_residual": 2.8586955394017954e-13,
    "donor_constraint_max_residual": 7.105427357601002e-15,
    "affine_coordinate_max_error": 4.440892098500626e-16,
    "quadratic_interpolation_relative_l2_errors": [
      0.0005430334768254892,
      0.00363893469369043
    ],
    "body_max_deviation_m": 6.497981000398245e-17,
    "component_relative_l2_errors": [
      0.001757311806124487,
      0.013373023323663687
    ],
    "physical_global_balance_relative_error": 0.0039805984762469925,
    "navier_stokes_verified": false,
    "strict_local_conservation": false
  }
}
```

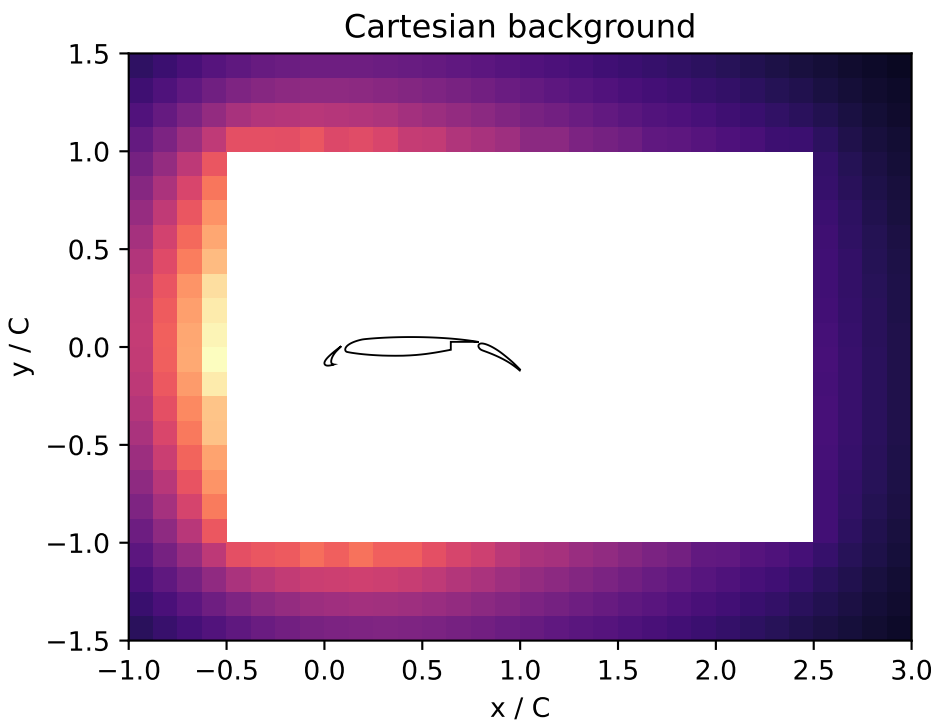
Background 64: mesh (physical-slot close-up)



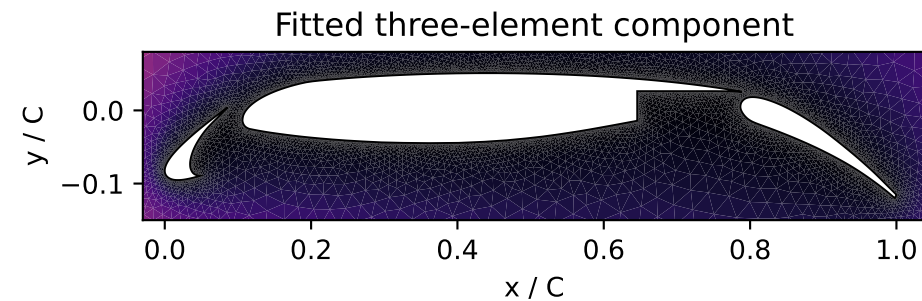
Background 64: solution (physical-slot close-up)



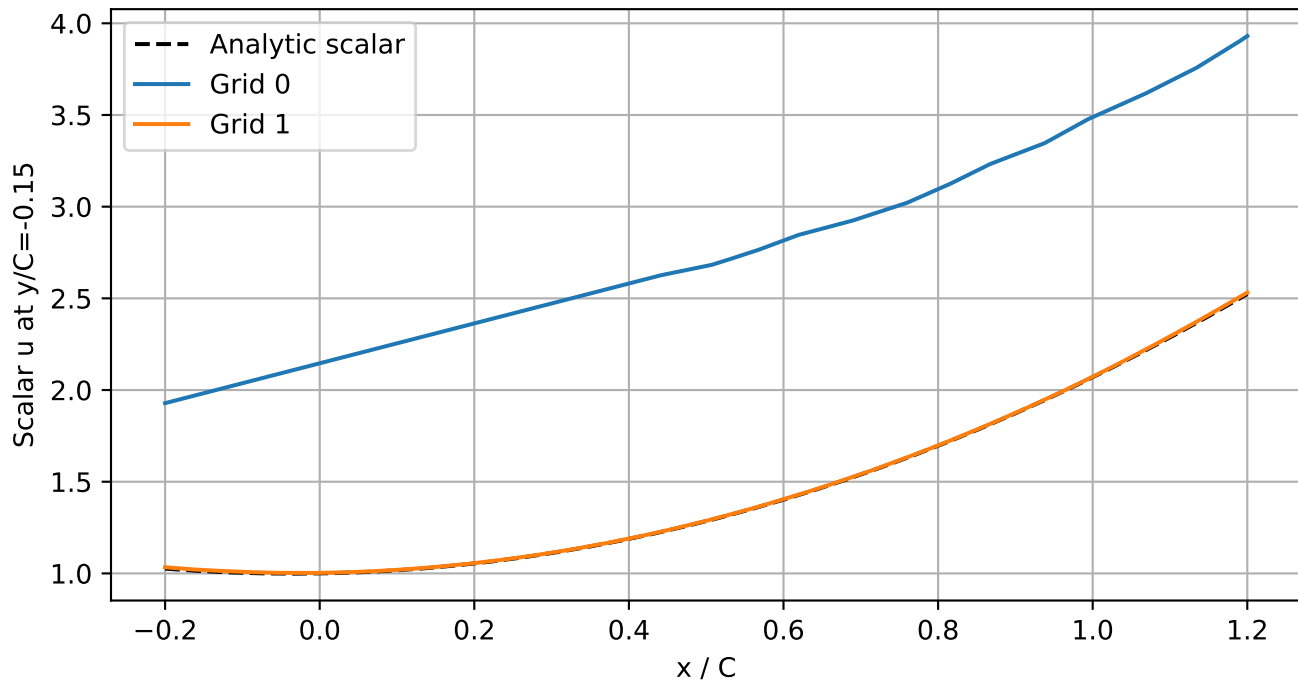
Background 64: error (physical-slot close-up)



Relative absolute scalar error



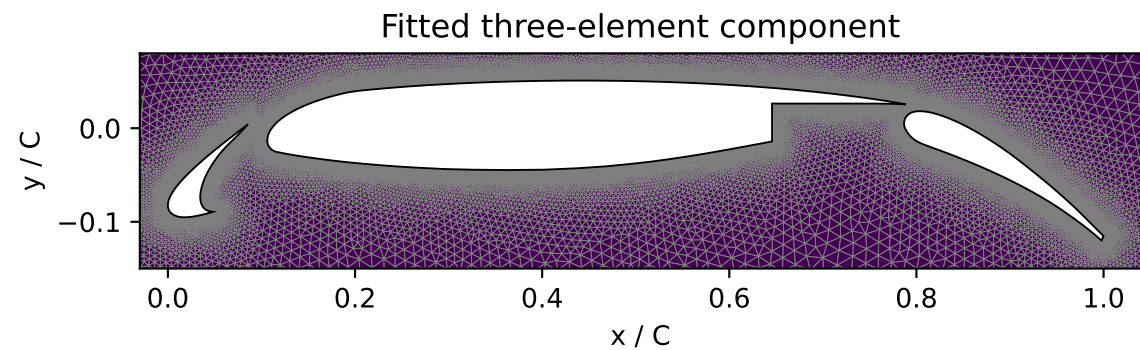
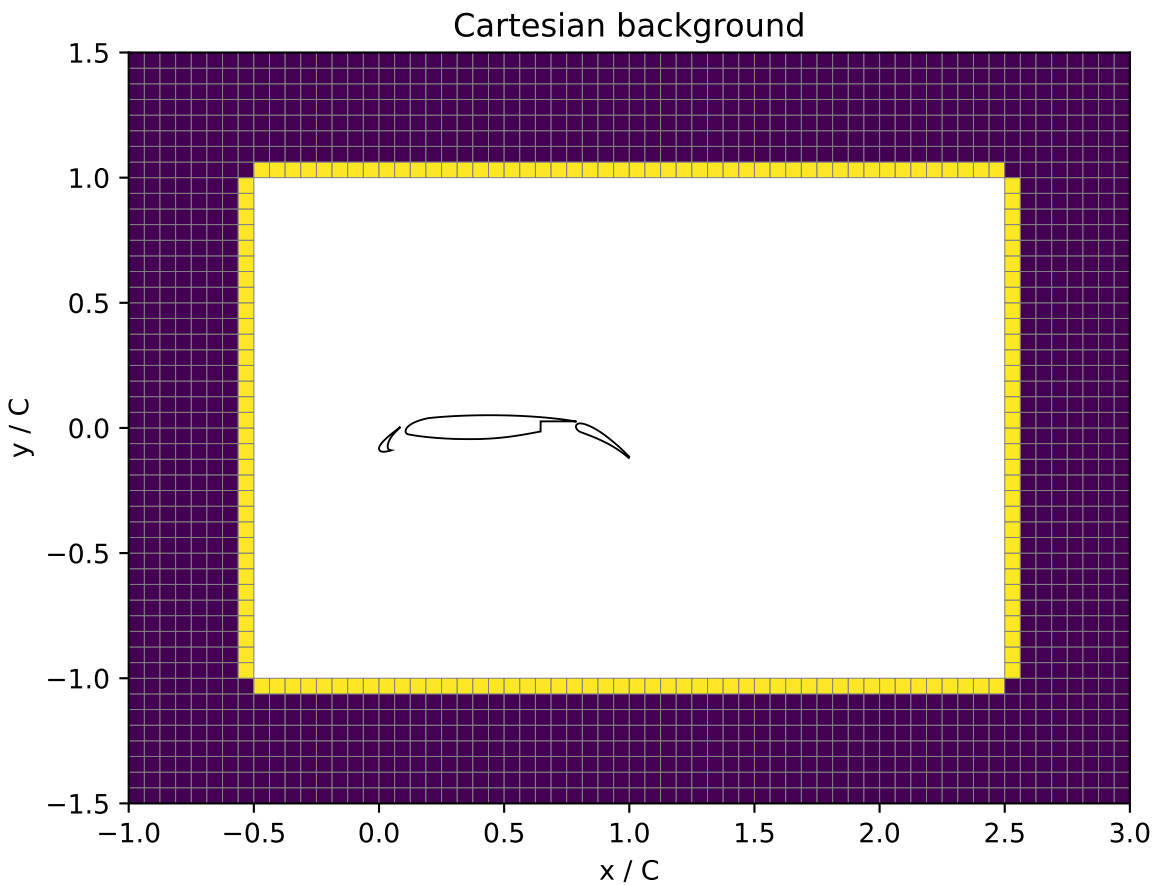
Relative absolute scalar error



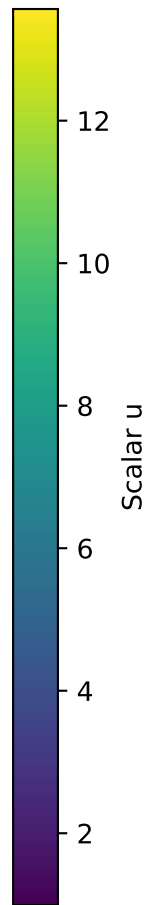
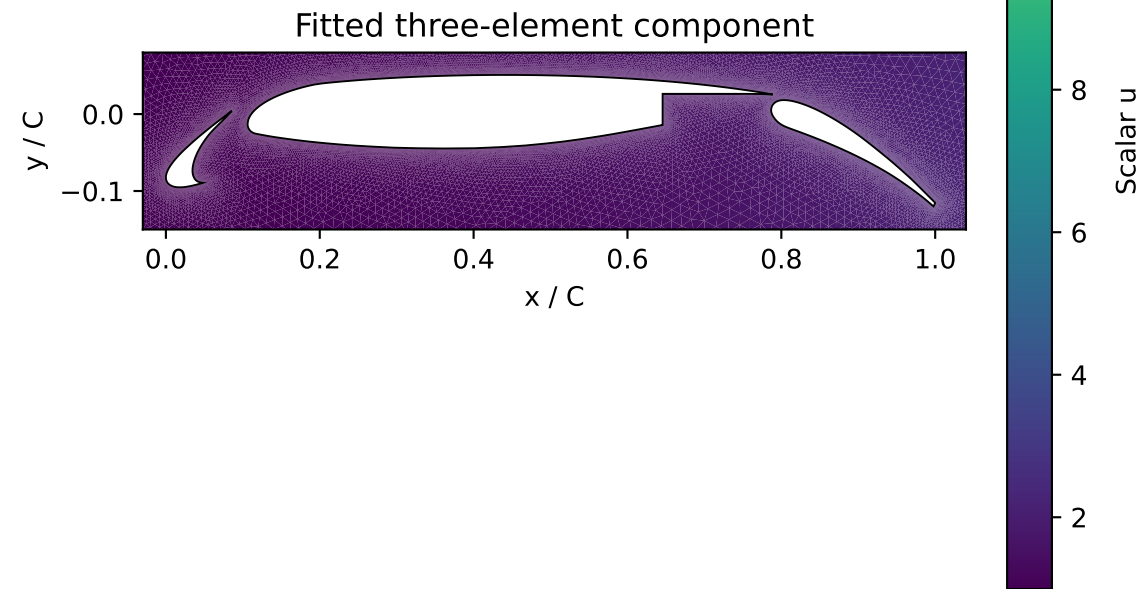
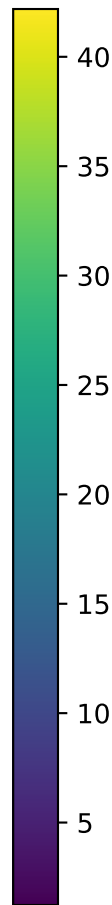
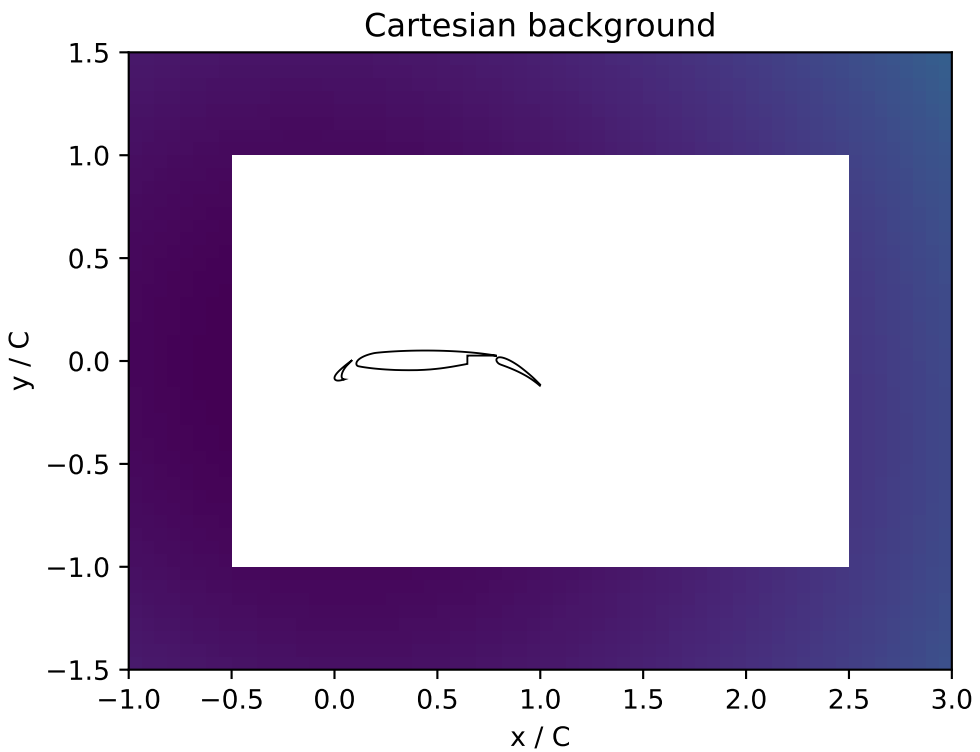
Grid 64: measured results

```
{
  "cells": [
    {
      "name": "background",
      "cells": 3584,
      "active": 3120,
      "hole": 384,
      "fringe": 80,
      "volume": 56.0
    },
    {
      "name": "30P30N fitted component",
      "cells": 30738,
      "active": 30504,
      "hole": 0,
      "fringe": 234,
      "volume": 11.9482836602939
    }
  ],
  "metrics": {
    "relative_algebraic_residual": 5.46762459468487e-16,
    "strict_local_conservation": false,
    "navier_stokes_verified": false,
    "grid_0_relative_l2_error": 0.0003686721134317284,
    "grid_1_relative_l2_error": 0.0027202613364316294,
    "maximum_donor_constraint_residual": 1.0658141036401503e-14,
    "physical_global_conservation_defect_relative": 0.0008146563242529423,
    "physical_boundary_diffusive_flux": -223.61082014871576,
    "physical_domain_integrated_source": -223.79313464117558,
    "active_cells": 33624,
    "fringe_cells": 314
  },
  "audit": {
    "resolution": 64,
    "passed": true,
    "independent_equation_max_residual": 2.783426158829705e-13,
    "donor_constraint_max_residual": 1.0658141036401503e-14,
    "affine_coordinate_max_error": 4.440892098500626e-16,
    "quadratic_interpolation_relative_l2_errors": [
      0.00015170775713016756,
      0.000895440070777384
    ],
    "body_max_deviation_m": 6.827306393595232e-17,
    "component_relative_l2_errors": [
      0.0003686721134317284,
      0.0027202613364316294
    ],
    "physical_global_balance_relative_error": 0.0008146563242534502,
    "navier_stokes_verified": false,
    "strict_local_conservation": false
  }
}
```

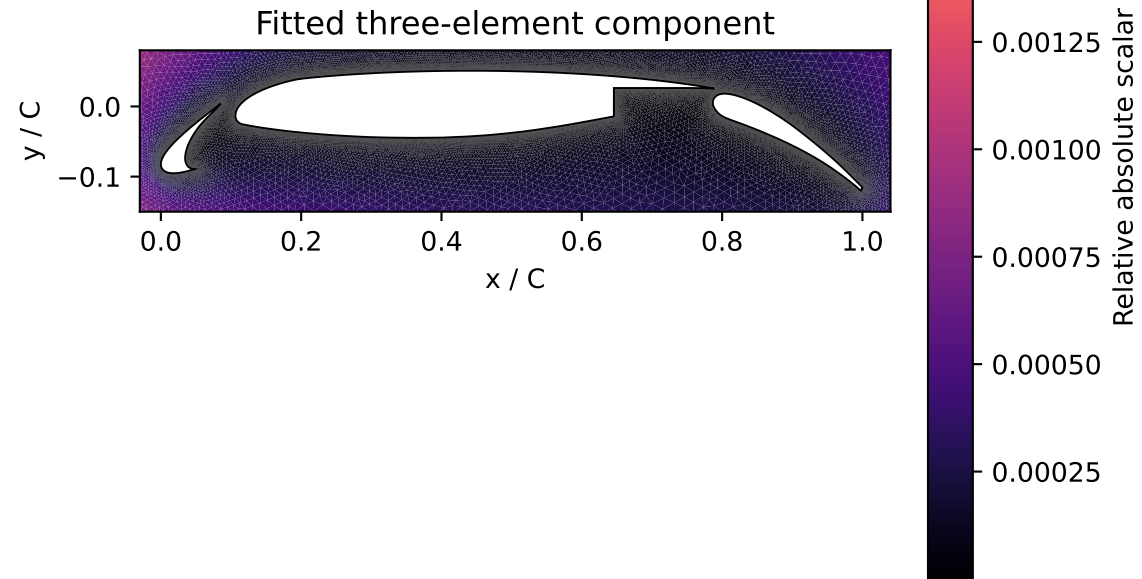
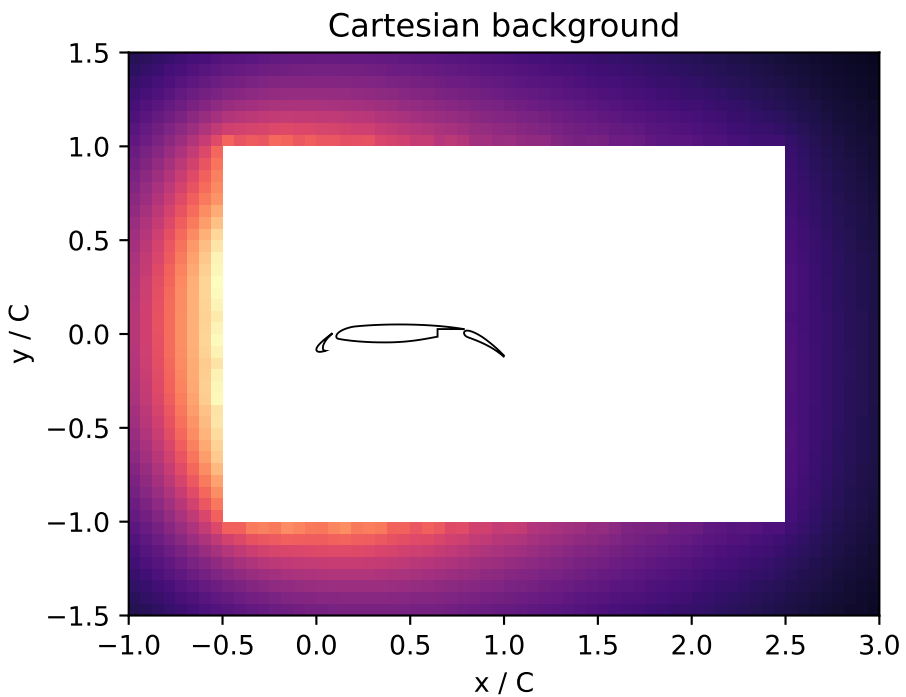
Background 128: mesh (physical-slot close-up)

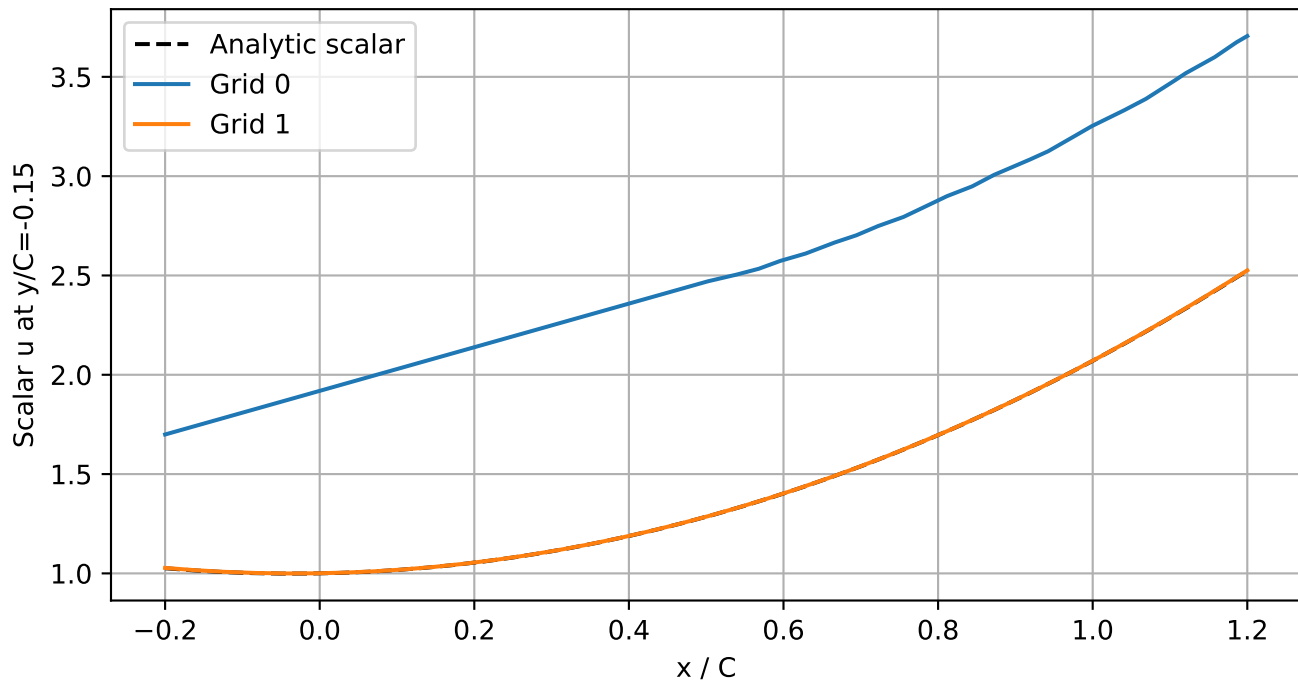


Background 128: solution (physical-slot close-up)



Background 128: error (physical-slot close-up)





Grid 128: measured results

```
{
  "cells": [
    {
      "name": "background",
      "cells": 14336,
      "active": 12640,
      "hole": 1536,
      "fringe": 160,
      "volume": 56.0
    },
    {
      "name": "30P30N fitted component",
      "cells": 95856,
      "active": 95388,
      "hole": 0,
      "fringe": 468,
      "volume": 11.948283660293898
    }
  ],
  "metrics": {
    "relative_algebraic_residual": 6.305805693810078e-16,
    "strict_local_conservation": false,
    "navier_stokes_verified": false,
    "grid_0_relative_l2_error": 8.6502177613767e-05,
    "grid_1_relative_l2_error": 0.0006271944144338529,
    "maximum_donor_constraint_residual": 2.4868995751603507e-14,
    "physical_global_conservation_defect_relative": 0.00018780649769704723,
    "physical_boundary_diffusive_flux": -223.75110483634998,
    "physical_domain_integrated_source": -223.79313464117558,
    "active_cells": 108028,
    "fringe_cells": 628
  },
  "audit": {
    "resolution": 128,
    "passed": true,
    "independent_equation_max_residual": 4.4452907067144e-13,
    "donor_constraint_max_residual": 2.4868995751603507e-14,
    "affine_coordinate_max_error": 4.440892098500626e-16,
    "quadratic_interpolation_relative_l2_errors": [
      3.873278786688868e-05,
      0.00022421527453209863
    ],
    "body_max_deviation_m": 7.392185903235177e-17,
    "component_relative_l2_errors": [
      8.6502177613767e-05,
      0.0006271944144338529
    ],
    "physical_global_balance_relative_error": 0.00018780649769882524,
    "navier_stokes_verified": false,
    "strict_local_conservation": false
  }
}
```

